5-MINUTE REVISION

Regression Analysis

From the chapter's own revision points, formulae and questions.

Selected from Chapter 9 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Regression → models the relationship between a dependent variable and one or more independent variables.
- Dependent variable → response or outcome to be predicted or explained.
- Independent variable → predictor or explanatory variable.
- Simple linear regression → one dependent variable and one independent variable.
- Multiple linear regression → one dependent variable and two or more independent variables.

Quick revision · 2

- Nonlinear regression → mean function nonlinear in its unknown parameters; polynomial regression can be linear in its coefficients.
- Logistic regression → categorical dependent variable.
- Line of best fit → fitted line obtained using least squares.
- Error → theoretical difference from the true regression relationship.
- Residual → observed value minus fitted value.

Quick revision · 3

- Least squares → minimizes the sum of squared residuals.
- Slope β → expected change in y for a one-unit increase in x.
- Intercept α → value of y when $x = 0$.
- Regression coefficient can range from $-\infty$ to $+\infty$.
- Positive regression coefficient → direct relationship.

Quick revision · 4

- Negative regression coefficient → inverse relationship.
- Regression of y on x → predict y from x.
- Regression of x on y → predict x from y.
- The regression line passes through $(\bar{x}, \bar{y})$.
- Both regression coefficients have the same sign.

Quick revision · 5

- $\beta_{yx}\beta_{xy} = r^2$.
- If $r = 0$, both regression coefficients are zero.
- If $r = \pm 1$, the two regression lines coincide.
- If one regression coefficient exceeds 1, the other must be less than 1.
- Both regression coefficients cannot exceed 1 simultaneously.

Quick revision · 6

- Regression coefficients are independent of change of origin but not of change of scale.
- Regression coefficients depend on the correlation and the relative standard deviations of the variables.
- Linearity, independence, homoscedasticity, normality of errors for exact inference, absence of exact multicollinearity, absence of error autocorrelation, and correct model specification are important assumptions.
- Regression lines are unique for a given dataset.
- Correlation measures association; regression models the relationship and allows prediction.

Quick revision · 7

- Correlation is symmetric; regression is asymmetric.
- Correlation coefficient is unit-free and lies between -1 and $+1$.
- Regression coefficients have units and are not bounded.
- Regression does not establish causation by itself.

Important formulae · 1

Straight-line equation:

$$Y = \alpha + \beta X$$

Simple linear regression model:

$$y_i = \alpha + \beta x_i + e_i$$

Residual after fitting:

$$\hat{e}_i = y_i - (\hat{\alpha} + \hat{\beta} x_i)$$

Important formulae · 2

Sum of squared errors:

$$E = \sum_{i=1}^{n} e_i^2$$

Sum of squared errors in terms of the regression model:

$$E = \sum_{i=1}^{n} [y_i - (\alpha + \beta x_i)]^2$$

First normal equation:

$$\sum_{i=1}^{n} y_i = n\alpha + \beta \sum_{i=1}^{n} x_i$$

Important formulae · 3

Second normal equation:

$$\sum_{i=1}^{n} x_i y_i = \alpha \sum_{i=1}^{n} x_i + \beta \sum_{i=1}^{n} x_i^2$$

Mean relationship:

$$\bar{y} = \alpha + \beta \bar{x}$$

Intercept:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Important formulae · 4

Regression coefficient of y on x:

$$\hat{\beta}_{yx} = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

Regression coefficient using covariance and variance:

$$\hat{\beta}_{yx} = \frac{cov(x,y)}{var(x)}$$

Regression equation of y on x:

$$\hat{y} = \hat{\alpha} + \hat{\beta}_{yx} x$$

Important formulae · 5

Regression coefficient of x on y:

$$\hat{\beta}_{xy} = \frac{cov(x,y)}{var(y)}$$

Regression equation of x on y:

$$\hat{x} = \hat{\alpha}_1 + \hat{\beta}_{xy} y$$

Relationship between regression coefficients and correlation:

$$\beta_{yx}\beta_{xy} = r_{xy}^2$$

Important formulae · 6

Correlation coefficient from regression coefficients:

$$|r_{xy}| = \sqrt{\beta_{yx}\beta_{xy}}$$

If both regression coefficients are positive:

$$r_{xy} = \sqrt{\beta_{yx}\beta_{xy}}$$

If both regression coefficients are negative:

$$r_{xy} = -\sqrt{\beta_{yx}\beta_{xy}}$$

Important formulae · 7

Regression coefficient in terms of correlation and standard deviations:

$$\beta_{yx} = r\frac{\sigma_y}{\sigma_x}$$

$$\beta_{xy} = r\frac{\sigma_x}{\sigma_y}$$

Change of origin and scale:

$$X = \frac{x-a}{h}, \qquad Y = \frac{y-b}{k}$$

Transformed regression coefficients:

$$\beta_{YX} = \frac{h}{k}\beta_{yx}$$

$$\beta_{XY} = \frac{k}{h}\beta_{xy}$$

Important formulae · 8

Multiple linear regression model:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + e$$

Self-check from the chapter

1. The variable predicted by a regression model is called the _____ variable. **[Dependent]**
2. A regression model with one predictor and a straight-line mean response is called _____ linear regression. **[Simple]**
3. The coefficient representing change in mean response per unit increase in the predictor is the _____. **[Slope]**
4. The predicted response when the predictor is zero is the _____. **[Intercept]**